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Search titled:

**The effectiveness of using artificial intelligence techniques in
developing social and linguistic communication
skills in autistic children.**

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Abstract.

To identify the effectiveness of using artificial intelligence techniques in developing social and language communication skills in children with autism according to the variables (gender and age).

To explore the effectiveness of using artificial intelligence techniques in developing social and language communication skills among autistic children, based on variables such as the student's academic year and school type. Identifying the impact of parental education on the development of social and language communication skills in children with autism using artificial intelligence techniques.

Study sample: 100 autistic children with mild symptoms, according to the students' school records & My mean was 4.32 and my standard deviation was 0.75.

They were randomly selected from Retaj School, the Intellectual Education School, Al-Yusr School for Special Needs, Results1-The findings of the study revealed that the use of artificial intelligence (AI) techniques significantly contributes to the development of social and linguistic communication skills among children with autism spectrum disorder (ASD).

The experimental group, which was exposed to AI-based interventions, showed considerable improvements compared to the control group, as indicated by statistical analysis using independent samples t-tests.2- Age played a significant role in the effectiveness of AI tools, with older children (10–12 years) demonstrating greater progress than younger ones (8–10 years), based on one-way ANOVA results.

However, no statistically significant differences were found between male and female participants, indicating that both genders benefit equally from AI-assisted learning. 3- School type also influenced outcomes, with students in private and international schools reporting higher levels of improvement due to better access to technology and trained personnel.

Parental education level did not show a significant impact on skill development; however, most parents reported being actively involved in supporting their children's use of AI at home. 4-Moreover, the questionnaire demonstrated high reliability across all three axes, with Cronbach's Alpha values exceeding 0.83, confirming its validity and consistency in measuring the intended constructs.

Keywords: Artificial Intelligence Technologies -Social Skills -Language Communication Skills - Autistic Children.

ملخص باللغة العربية.

فاعلية استخدام تقنيات الذكاء الاصطناعي في تنمية المهارات الاجتماعية ومهارات التواصل اللغوي لدى الأطفال من ذوي اضطراب التوحد.

هدفت الدراسة إلى: التعرف على فعالية استخدام تقنيات الذكاء الاصطناعي في تنمية مهارات التواصل الاجتماعي واللغوي لدى الأطفال المصابين بالتوحد، بناءً على متغيرات مثل السنة الدراسية للطالب ونوع المدرسة.

والتعرف على أثر التربية الوالدية على تنمية مهارات التواصل الاجتماعي واللغوي لدى الأطفال المصابين بالتوحد باستخدام تقنيات الذكاء الاصطناعي. عينة الدراسة: ١٠٠ طفل مصاب بالتوحد بأعراض خفيفة، حسب السجلات المدرسية للطلبة بنسب وكان متوسط حسابي ٤,٣٢ وانحراف معياري ٠,٧٥. وقد تم اختيارهم عشوائياً من مدرسة ريتاج، ومدرسة التربية الفكرية، ومدرسة اليسر لذوي الاحتياجات الخاصة، النتائج.

و كشفت نتائج الدراسة أن استخدام تقنيات الذكاء الاصطناعي يساهم بشكل كبير في تنمية مهارات التواصل الاجتماعي واللغوي لدى الأطفال ذوي اضطراب طيف التوحد وأظهرت المجموعة التجريبية، التي تعرضت لتدخلات تعتمد على الذكاء الاصطناعي، تحسنات كبيرة

مقارنة بالمجموعة الضابطة، كما أشارت التحليلات الإحصائية باستخدام اختبارات t للعينات المستقلة.

ولعب العمر دورًا مهمًا في فعالية أدوات الذكاء الاصطناعي، حيث أظهر الأطفال الأكبر سنًا (١٠-١٢ عامًا) تقدمًا أكبر من الأطفال الأصغر سنًا (٨-١٠ سنوات)، بناءً على نتائج تحليل التباين أحادي الاتجاه.

ومع ذلك، لم يتم العثور على فروق ذات دلالة إحصائية بين المشاركين الذكور والإناث، مما يشير إلى أن كلا الجنسين يستفيدان بالتساوي من التعلم بمساعدة الذكاء الاصطناعي.

كما أثر نوع المدرسة على النتائج، حيث أفاد الطلاب في المدارس الخاصة والدولية بمستويات أعلى من التحسن بسبب سهولة الوصول إلى التكنولوجيا والموظفين المدربين. ولم يظهر مستوى تعليم الوالدين تأثيرًا كبيرًا على تنمية المهارات؛ ومع ذلك، أفاد معظم الآباء بأنهم يشاركون بشكل نشط في دعم استخدام أطفالهم للذكاء الاصطناعي في المنزل.

كما أظهرت الاستبانة ثباتًا عاليًا على جميع المحاور الثلاثة، حيث تجاوزت قيم ألفا كرونباخ ٠,٨٣، مما يؤكد صلاحيتها وثباتها في قياس المفاهيم المقصودة.

الكلمات المفتاحية: (تقنيات الذكاء الاصطناعي- المهارات الاجتماعية- مهارات التواصل اللغوي- الأطفال التوحدين).

The effectiveness of using artificial intelligence techniques in developing social and linguistic communication skills in autistic children.

introduction:

Autism spectrum disorder have social and emotional needs that impact their academic performance and are a the foundation of social and emotional competence and one of the greatest deficits in how children with autism socialize (Christine K Syriopoulou-Delli , Eleni Gkiolnta ,Kyriaki Sarri ,Areti- Eirini Filiou &Elpis Papaefstathiou ,2024).

Difficulties associated with this type of communication vary widely and include understanding facial expressions, using conversational gestures, imitating body movements, and maintaining joint attention. (Legg H, Tickle A.,2019).

Research has indicated that early deficits in these skills lead to later difficulties in acquiring more complex social skills. Emotions conveyed through facial expressions, gestures, words, or postures are important cues for understanding the feelings and intentions of others and regulating social interactions. (Goodwin MS,2018).

Therefore, it is essential that schools provide interventions that address these needs, using appropriate tools.(Howlin P, Moss P, Savage S, Rutter M.,2013).

For nearly two decades, the term “autism spectrum disorders” has been used instead of “early childhood autism” to refer to disorders that affect individuals who share common behavioral and cognitive features, but to varying degrees, and are primarily related to a specific underlying brain condition/factor. (Khodatars M, Shoeibi A, Sadeghi D, Ghaasemi N, Jafari M, Moridian P, Khadem A, Alizadehsani R, Zare A, Kong Y, et al.,2021).

For the latter reason, these disorders were included in the Diagnostic and Statistical Manual of Mental Disorders (DSM-IV) as an autistic disorder, as opposed to the environmental/familial factor that was believed for many years to be the predominant factor in the onset and persistence of autism.

It is also classified as a disorder that arises “from birth or within the first 30 months of life” and is characterized by: a lack of emotional connection with others; excessive attachment to routines, which may sometimes include eccentricity; speech or mutism; fascination with and skill in handling certain objects; and higher levels of visuospatial skills and memory. (Belpaeme T, Baxter PE, Read R, Wood R, Cuayáhuatl H, Kiefer B, Racioppa S, Kruijff-Korabayová I, Athanasopoulos G, Enescu V, et al., 2013).

Study Problem:

Autism Spectrum Disorder (ASD) has become one of the most prevalent neurodevelopmental disorders globally. According to the Centers for Disease Control and Prevention (CDC, 2023), approximately 1 in every 54

children is diagnosed with ASD in the United States.

Meanwhile, recent local reports from Saudi Arabia (Ministry of Health, 2021) indicate that the prevalence rate in some Arab countries reaches about 1.7% among children , highlighting the increasing need for effective intervention strategies tailored to this growing population.

Children with ASD often face significant challenges in social interaction , emotional regulation , and language communication , which affect their ability to engage effectively in educational and social environments (Christine et al., 2024).

These difficulties include trouble understanding facial expressions, maintaining eye contact, using gestures, initiating conversations, and responding appropriately in social contexts. As a result, they require structured, repetitive, and personalized interventions that address these deficits systematically.

Traditional therapeutic approaches such as Applied Behavior Analysis (ABA) have been widely used to improve the skills of children with ASD. However, these methods often suffer from limitations including high costs, limited availability of trained professionals, and inconsistency in implementation (Howlin et al., 2013; Du et al., 2024).

Additionally, many of these interventions are not easily adaptable to individual learning styles or cultural contexts, especially in Arabic-speaking communities.

In response to these challenges, Artificial Intelligence (AI)-based technologies have emerged as promising tools for delivering more consistent, engaging, and personalized interventions. AI applications such as speech recognition software, virtual reality platforms, robotic systems, and adaptive learning apps provide interactive environments that can simulate real-life scenarios and offer immediate feedback - essential components for improving social and language skills in children with ASD (Kotsi et al., 2025; Belpaeme et al., 2013).

Several studies have demonstrated the effectiveness of AI-based interventions in enhancing the communication abilities of children with autism. For example, Alves et al. (2020) found that assistive technologies powered by AI significantly improved behavioral and communicative outcomes among autistic children.

Similarly, Chuan-Po Wang (2023) reported that AI-powered robots helped children with ASD organize sentence structures and interact more effectively in guided communication settings.

Despite these promising findings, there remains a gap in the literature regarding the application of AI technologies in Arabic-speaking contexts. Most existing studies focus on Western populations and may not fully account for cultural, linguistic, or educational differences that could influence the effectiveness of AI-based interventions. Furthermore, few studies have explored how variables such as age, school type, and parental involvement moderate the impact of AI on skill development among children with ASD in Middle Eastern or North African (MENA) regions.

This study addresses this gap by examining the effectiveness of artificial intelligence techniques in developing social and linguistic communication skills among children with ASD in Egypt , while also investigating how factors such as age, gender, school type, and parental education influence the outcomes.

It aims to contribute both theoretically and practically by validating the use of AI-based tools within an Arabic educational framework and providing evidence-based recommendations for educators, therapists, and policymakers.

Study objectives:

1. Main objective: To identify the effectiveness of using artificial intelligence techniques in developing social and language communication skills in children with autism.
2. To identify the effectiveness of using artificial intelligence techniques in developing social and language communication skills in children with autism according to the variables (gender and age).

3. To explore the effectiveness of using artificial intelligence techniques in developing social and language communication skills among autistic children, based on variables such as the student's academic year and school type.
4. Identifying the impact of parental education on the development of social and language communication skills in children with autism using artificial intelligence techniques.

Study Questions:

1. Main Question: What is the effectiveness of using artificial intelligence techniques in developing social and language communication skills in autistic children?
2. What is the effectiveness of using artificial intelligence techniques in developing social and language communication skills in autistic children according to the variables (gender and age)?

3. What is the impact of using artificial intelligence technologies on the development of linguistic communication between students after using artificial intelligence technologies?
4. Is there an impact of parental education on the development of social and language communication skills in autistic children using artificial intelligence techniques?

Study hypotheses:

- 1) There are no statistically significant differences in the development of social and communication skills between children in the experimental group and the control group after using artificial intelligence techniques.
- 2) There are statistically significant differences in favor of the experimental group in social and communication skills after using artificial intelligence techniques.
- 3) There are statistically significant differences in the use of artificial intelligence techniques in developing social and communication skills among autistic children according to the variables (gender and age).

- 4) There are no statistically significant differences in the effect of parental education on the development of social and linguistic communication skills in children with autism using artificial intelligence techniques.

Importance of the study:

Theoretical importance:

- 1) One approach to try to establish a bridge between the person with ASD and their immediate world lies in seeking strategies mediated by augmentative and alternative communication systems.

This refers to the implementation of technologies emerging from therapeutic clinical practice as tools to help resolve the communication challenges that many people with ASD experience, helping to compensate for these deficiencies.

- 2) In recent years, AAC devices have leveraged digital technologies (computers, cell phones, digital tablets, etc.) with software that

offers multiple functions, is accessible, and offers rapid response times, allowing the barrier posed by communication difficulties to be less rigid; this constitutes the AAC device itself .

Practical importance:

- ✓ Produces results that benefit scientific research, as artificial intelligence is applied to various fields and problems, including education and the social welfare of students in general.
- ✓ One emerging application of AI is to help people with autism, a developmental condition that affects how people communicate and interact with others, as well as their interests and behaviors.

Study terms:

Artificial intelligence techniques.

- ✓ **Technical definition:** There are many types of machine learning techniques or algorithms. Each of these approaches is suitable for differ-

ent types of problems and data.

La forma más sencilla de machine learning es llamada aprendizaje supervisado, que implica el uso de conjuntos de datos etiquetados para capacitar algoritmos a clasificar datos o predecir resultados con precisión. En el aprendizaje supervisado, los humanos emparejan cada ejemplo de entrenamiento con una etiqueta de salida.

El objetivo es que el modelo aprenda el mapeo entre entradas y salidas en los datos de entrenamiento, para que pueda predecir las etiquetas de datos nuevos e invisibles. (Sheikh, H., Prins, C., Schrijvers, E. (2023).

✓ **Operational definition** :is a technology that enables computers and machines to simulate human learning, understanding, problem-solving, decision-making, creativity, and autonomy.AI-equipped applications and devices can see and identify objects.

They can understand and respond to human language. They can learn from new information and experience. They can make detailed recommendations to users and experts. They can act independently, replacing the need for human intelligence or intervention (a classic example is a self-driving car).

Social skills.

✓ **The technical definition:** is as follows: they are competencies that allow us to communicate, relate, and interact with others effectively and appropriately. These skills are important for establishing and maintaining healthy interpersonal relationships and for adapting to different contexts and cultures (Du, G., Guo, Y. & Xu, W. ,2024).

✓ **Operational definition:** are the set of behaviors that allow human beings to maintain positive interpersonal relationships with others and respond appropriately to the demands of their environment. When an individual uses these skills appropriately, a mutually satisfying situation arises, which acts as positive reinforcement. There are numerous defini-

tions for social skills, as they are a complex concept. Furthermore, they depend on the cultural context in which they are manifested, so the same characteristic may be valued positively in one culture and negatively in another.

Linguistic communication skills.

✓ **Technical definition:** The communication and information communication between other people. There are many different forms among people with different information, including verbal language, writing and/or no verbal (corporal) communication. The company is able to communicate with Brendan the capacity to transmit the effective management and efficient sus pens and ideas (Fathi, Syed Ahmad ,2023)

✓ **Operational definition:** involve interacting orally, in writing, with sign language, or in a multimodal manner in a coherent and appropriate manner in different settings and contexts, and for different communicative purposes.

Previous studies:

- ✓ Al Sharkawy, S. M., & Al Hattalia, A. S. (2024). Effectiveness of Artificial Intelligence Applications in Developing Expressive Language Skills in Children with Autism Spectrum Disorder (Application of a Skill as a Model).

The study sample consisted of 10 children with autism spectrum disorder, aged between 5-7 years, attending a rehabilitation center in the state of Nizwa, The results also showed the continuity of this improvement during the follow-up period.

The study concluded the importance of adopting this strategy in other intervention programs aimed at enhancing the speech abilities of children with autism spectrum disorder and reducing their common language problems.

Commentary on previous studies:

In previous studies, the researcher reviewed previous studies on the study's variables and linked them to each other in order to benefit from previous scientific content and rely on it in writing scientific content.

The researcher benefited from presenting some previous studies, although there is a common factor between the previous studies that were reviewed in studying the various factors that may affect the language communication skills of autistic children.

Some previous Arab studies addressed the factors that affected the language communication skills of autistic children. The researcher was interested in reviewing previous foreign studies, where he found many points that focused on the necessity of paying attention to the language communication skills of autistic children. There was great importance in developing the language communication skills of the autistic children under study.

Theoretical framework:

History of the Onset of the Disorder.

Hans Asperger's (1944) discovery of individuals who exhibited these characteristics, but to varying degrees, and whom he classified under the name "Asperger's Syndrome," highlighted the possibility of a spectrum of symptoms that are fundamentally similar, rather than specific.

Asperger's Syndrome exhibited the following: an inappropriate and naive social approach; limited interests in certain subjects or objects; correct grammar and vocabulary, but poor use of it in conversation; a lack of common sense; impaired motor coordination; learning difficulties in one or two areas, and a higher level of voluntary achievement in others (Kotsi, S., Handrinou, S., Iatraki, G., & Soulis, S.-G. ,2025).

This description showed similarities to Kanner's description of autism, but it was impossible to consider them identical due to their marked differences in degree, particularly in language (whose impaired subdomain in

this case was pragmatics/conversational, not phonology, morphology, and/or semantics as in other autistic people), and social relationships (which in Asperger's appear only with a lack of 'common sense' or understanding of 'shared symbols', rather than with a more pronounced isolation or apathy as in other autistic people). (Li, G., Zarei, M. A., Alibakhshi, G., & Labbafi, A., 2024).

For this reason, it became increasingly necessary to change how we view the different but shared characteristics of people with autism. Research by various specialists in the 1970s and 1980s supported this trend, so that the term "autism spectrum" was used in new classification manuals and in the considerations of new studies on the subject. (Khodatars M, Shoeibi A, Sadeghi D, Ghaasemi N, Jafari M, Moridian P, Khadem A, Alizadehsani R, Zare A, Kong Y, et al., 2021).

Diagnostic and Statistical Manual of Mental Disorders (DSM-5).

The common basis of symptoms is currently established as follows: impairment in social relationships, in communication, and in flexibility of interests and imagination. On this basis, the autism spectrum (within the Kanner and Asperger spectrums).

lies (Posar, A., Resca, F., & Visconti, P. ,2015) The addition of equally important, more “cognitive” characteristics further highlights the importance of an autism diagnosis: These include difficulty in consolidating previous information and an inability to categorize experiences in time (past, present, future) and space; difficulty in selecting key information from the secondary information that arrives.

The poverty of discovering the feelings or needs of another person, i.e. in the ability of “reason”, and the metaphorical or indirect meaning of messages; an inexperienced ability to generalize learning acquired in one environment to another; and a deficit in working memory (retaining different types of information for short periods of time while simultaneously us-

ing them) (Druga S, Vu ST, Likhith E, Qiu T. ,2019).

Autistic people also suffer from a lack of attention due to poor visual coordination, which makes them unable to imitate others, leading to apparent deficits in language, play, and social development.(Ola I, (2011)

In addition to the motor difficulties experienced by many people with autism, there is also abnormal sensitivity (to a greater extent - hypersensitivity or to a lesser extent - hyposensitivity) in one or more of the senses (touch, sight, hearing, taste, and smell).

1441 More common in people outside the autism spectrum are tantrums, aggression, self-harm, anxiety, tics, hyperactivity, eating, sleeping, toilet training, and self-care problems; many of which are the result of deficits in communication, perception, and sensory processing. (M. B. Colton, D. J. Ricks, M. A. Goodrich, B. Dariush, K. Fujimura, M. Fujiki, ,2009).

Focusing on the language skills of autistic people, we find that their language develops slowly and may not develop at all, They may repeat some words without realizing their meaning, and may repeat words with same rhythm in what is known (echolalia).

They also use pronouns in reverse. Most autistic don't express what they need in words, but rather in gestures. They rarely use words that indicate mental abilities, such as (remember), (believe). (Ola A, (2015), (Iman F, 2017).

In short, the skills that are deficient and excessive in autism, and which can be increased, improved, and modified through treatment once a comprehensive diagnosis is made, are diverse and can reach a large number. Currently, there has been an increase in scientific production to identify the causes of autism spectrum disorder, as well as a compilation of the development of treatments. (Estepan PG, Baxter P, Belpaeme T, Billing E, Cai H, Cao H-L, Coeckelbergh M, Costescu C, David D, De Beir A, et al. ,2017).

Therapy.

Considering therapy helps overcome the overwhelming need to categorize a person as a subset of the same group, which, moreover, can develop into another group over time and often exhibits traits of others.

While some groups of traits are more likely than others, their boundaries are difficult to define. What is crucial from a clinical perspective is the ability to diagnose whether a person belongs to the spectrum (Foxx R. M., 2008).

1441 This is where empirically supported treatments are successfully applied. Currently the treatments with the greatest validity due to their clinical effectiveness in the autism spectrum (Treatment and Education for Children with Autism-Associated Communication Difficulties). (Gitimoghaddam, M., Chichkine, N., McArthur, L., Sangha, S. S., & Symington, V., 2022).

The primary method in ABA therapy for teaching skills (cognitive, language, academic, motor, behavioral, social, and play) is structured, discrete-task-based learning.

This includes: breaking the skill into small parts, teaching one part at a time, with a beginning marked by a stimulus (some kind of instruction, a model, physical or verbal prompting, or cues), and an end marked by a reinforcer (which can be, among other things, different depending on the performance of the skill part; e.g., if achieved by prompting, it is reinforced in one way, and if achieved independently, in another way), or its absence (increasing information that the response was incorrect or missing), in case the expected response does not occur. (Du, G., Guo, Y. & Xu, W. ,2024).

The basic requirement is that the second part of the skill cannot be moved on until the first part is fully achieved, subject to a pre-agreed mastery criterion (usually when the sub-skill has been performed correctly and independently for several consecutive days).

This includes, as an intermediate step and depending on the situation, reinforcement by bringing the required part of the skill closer (or shaping it), i.e. towards a response similar, to varying degrees and after prior agreement, to the response intended to be achieved before moving on to the second skill.

The second condition is the intermingling of leisure activities, whether free or used to benefit the training of specific skills (increasing cooperative play, increasing the functional use of objects, reducing tantrums), without the structure imposed by discriminatory tasks.

The goal is not to automate learning, but rather to naturalize it as much as possible and embody it in stimulating activities. This is not to forget that people with ASD, due to the state of their brains and information processing, require a greater learning structure and number of repetitions when training any skill, in contrast to learning through continuous tasks, which is common in the formal education system, and passive.

Incidental, verbal learning, which is more or less sufficient for people without ASD. The third condition is to achieve gradual generalization

of learning across different skills, which requires a more flexible structure.

(F. J. Alves, E. A. De Carvalho, J. Aguilar, L. L. De Brito and G. S. Bastos, 2020)

This is achieved by addressing various aspects. First, an attempt is made to remove stimuli (stimuli, models, etc.) and, at the same time, modify them (more complex and varied instructions; given by different people).

Second, changing the reinforcers, both their type (artificial rather than natural; i.e., social or self-reinforcing reinforcement) and the program on which they are based (ratio or interval, fixed for variables; i.e., less and less artificial reinforcement), leads to incidental rather than systematic learning.

Finally, the transition to other learning environments, as natural as possible (home, school, workshop), which subsequently leads to other people carrying it out (parents, relatives, teachers, peers). (Bottema-Beutel K and Crowley S, 2021) .

Methodological procedures:

- ✓ Study population: Cairo.
- ✓ Study sample: 100 autistic children with mild symptoms, according to the students' school records. They were randomly selected from Retaj School, the Intellectual Education School, Al-Yusr School for Special Needs.
- ✓ Study methodology: Quasi-experimental.
- ✓ Study boundaries:
 - Spatial boundary: Cairo.
 - Temporal boundary: During 2025 AD / 1446 AH.
 - Human boundary: 100 autistic children with mild symptoms, according to the students' school records, who were randomly selected.

Statistical Analysis.

Methodological procedures:

- Study population: Cairo.

- Study sample: 100 autistic children with mild symptoms, according to the students' school records. They were randomly selected from Retaj School - Fifth Settlement, the Intellectual Education School, Al-Yusr School for Special Needs, and the.
- Study methodology: Quasi-experimental.
- Study boundaries:
 - Spatial boundary: Cairo.
 - Temporal boundary: During 2025 AD / 1446 AH.
 - Human boundary: 100 autistic children with mild symptoms, according to the students' school records, who were randomly selected.

Study hypotheses:

- 1) There are no statistically significant differences in the development of social and communication skills between children in the experimental group and the control group after using artificial intelligence techniques.

2) There are statistically significant differences in favor of the experimental group in social and communication skills after using artificial intelligence techniques.

3) There are statistically significant differences in the use of artificial intelligence techniques in developing social and communication skills among autistic children according to the variables (gender and age).

4) There are no statistically significant differences in the effect of parental education on the development of social and linguistic communication skills in children with autism using artificial intelligence techniques.

5)

2. Demographic Distribution of the Sample.

Table 1: Frequency and Percentage Distribution of the Sample According to Age Groups.

Age Group	Frequency	Percentage
8–10 years	63	63%
10–12 years	37	37%
	100	100%

The sample distribution according to age groups reveals that the majority of participants (63%) fall within the 8–10 years age range, while a smaller proportion (37%) are aged 10–12 years.

This unequal distribution suggests that younger children with mild autism symptoms are more commonly represented in the selected schools. The predominance of the younger age group may influence the generalization of results across all age ranges within the targeted developmental stages. However, it still allows for meaningful comparison and analysis of the impact of AI-based interventions on different age levels.

Table 2: Gender Distribution of the Sample.

Gender	Frequency	Percentage
Male	70	70%
Female	30	30%

There is a higher proportion of male participants compared to female participants, which aligns with epidemiological trends indicating that ASD

is more commonly diagnosed in males. However, this imbalance should be considered when interpreting gender-related findings.

Table 3: Distribution of Participants by Type of School.

School Type	Frequency	Percentage
Government	40	40%
Private	45	45%
International	15	15%

The majority of participants attended either government or private schools, with fewer enrolled in international schools. This distribution reflects the accessibility of AI-based interventions within different educational systems in Cairo.

Table 4: Parents' Educational Level.

Parental Education Level	Frequency	Percentage
Illiterate	10	10%
Intermediate	20	20%
University	50	50%
Postgraduate	20	20%

Most parents had at least a university-level education, which may positively influence their ability to support AI-assisted learning at home. Only 10% of parents were illiterate, which could affect the consistency of home-based reinforcement of AI tools.

3. Hypotheses Testing

Hypothesis 1:

There are no statistically significant differences in the development of social and communication skills between children in the experimental group and the control group after using artificial intelligence techniques.

Table 5: Comparison of Social and Communication Skills Between Experimental and Control Groups Using Independent Samples t-test.

Group	Mean Score	Std. Deviation	t-value	Sig. (2-tailed)
Experimental	4.32	0.75	5.14	0.001*
Control	3.15	0.89		

Note: * Significant at $p < 0.05$.

The results show a statistically significant difference between the experimental and control groups, with the experimental group achieving higher mean scores in social and communication skills. Therefore, we reject the null hypothesis. This indicates that AI techniques significantly enhance these skills in autistic children.

Hypothesis 2:

There are statistically significant differences in favor of the experimental group in social and communication skills after using artificial intelligence techniques.

As shown in Table 5 above, the experimental group scored significantly higher than the control group ($p = 0.001$). Thus, this hypothesis is supported by the data.

Hypothesis 3:

There are statistically significant differences in the use of artificial intelligence techniques in developing social and communication skills among autistic children according to the variables (gender and age).

Table 6: One-Way ANOVA Results for Differences

Based on Age and Gender.

Variable	F-value	Sig. (p-value)	Conclusion
Age	4.89	0.028*	Significant
Gender	1.24	0.267	Not significant

Note: * Significant at $p < 0.05$.

The ANOVA results indicate a statistically significant effect of age on the development of social and communication skills using AI techniques.

Older children (10–12 years) showed greater improvement than younger ones (8–10 years). However, there was no significant difference based on gender, suggesting that both males and females benefit similarly from AI interventions.

Hypothesis 4:

There are no statistically significant differences in the effect of parental education on the development of social and linguistic communication skills in children with autism using artificial intelligence techniques.

Table 7: Chi-Square Test of Independence Between

Parental Education and Skill Development.

	χ^2 value	df	p-value	Conclusion
Parental Education Level	7.34	3	0.062	Not statistically significant

Although there is a trend toward significance ($p = 0.062$), it does not reach the conventional threshold of 0.05. Therefore, we accept the null hypothesis, indicating that parental education level did not have a statistically significant impact on skill development in this study.

4. Item Analysis (Selected Items).

1. Axis 1: Effectiveness of AI Techniques in Developing Social and Linguistic Communication Skills According to Gender and Age.

Table 7: Responses to Items Related to Axis 1 (Gender and Age).

Nu.	Statement	Agree (%)	Some-what Agree (%)	Disagree (%)
1	AI technology helped me increase my social interaction.	65%	25%	10%
2	AI technology helped me interact with peers.	70%	20%	10%
3	AI technology helped me interact with the teacher.	68%	22%	10%
4	AI technology helped me interact in educational activities.	72%	20%	8%
5	The older the child, the more new linguistic learning a child has.	60%	25%	15%
6	AI had a negative impact on children's learning of new linguistics.	10%	15%	75%
7	AI caused an inappropriate development in social communication skills.	5%	10%	85%

Nu.	Statement	Agree (%)	Some-what Agree (%)	Disagree (%)
8	Students' social communication skills did not develop significantly.	10%	15%	75%
9	Males are less affected by AI.	20%	25%	55%
10	Females are less affected by AI.	15%	20%	65%
11	Males are more affected by AI.	50%	30%	20%
12	Females are more affected by AI.	55%	30%	15%
13	Age has an impact on children's exposure to AI.	70%	20%	10%
14	There is no difference between males and females in their exposure to AI technologies.	20%	25%	55%

The majority of participants agreed that AI positively influenced their ability to engage socially with teachers, peers, and in classroom activities. A significant portion also believed that older children benefit more from AI in terms of language acquisition.

Negative perceptions were minimal, with most disagreeing that AI harms linguistic or social development. Regarding gender, slightly more agreement was found for females being more affected by AI, though this difference was not statistically significant. Most respondents agreed that age plays a role in how children interact with AI tools.

2. Axis 2: Effectiveness of AI Techniques Based on Academic Year and Type of School.

Table 8: Responses to Items Related to Axis 2

(Academic Year and School Type).

Nu.	Statement	Agree (%)	Some-what Agree (%)	Disagree (%)
15	The academic year makes no difference in students' responses .to AI techniques	20%	25%	55%
16	The academic year makes no difference in students' responses .to AI techniques	22%	23%	55%
17	The type of school makes no difference .in the use of AI	25%	20%	55%
18	The type of school makes a difference in .the use of AI	65%	25%	10%
19	Students benefited .from AI techniques	80%	15%	5%
20	Students did not benefit from AI tech- niques	5%	10%	85%
21	Students' develop- ment is based on the development of the teacher's abilities	40%	30%	30%
22	The language score increased in students' weekly assessments	70%	20%	10%

Nu.	Statement	Agree (%)	Some-what Agree (%)	Disagree (%)
23	The language score decreased in students' weekly assessments.	5%	10%	85%
24	Students' skills can be developed using traditional educational methods.	30%	35%	35%
25	Students' development is limited using AI techniques.	10%	15%	75%
26	Students' computer use improved after student training.	75%	20%	5%
27	Students were unable to use the computer.	5%	10%	85%
28	The researcher found that students' use of the computer was easy.	80%	15%	5%

There was strong agreement that students benefited from AI techniques, with notable improvements in language scores and computer use. Most disagreed that AI limits development or that academic year has no effect.

The majority believed that the type of school influences the effectiveness of AI implementation, with private and international schools having better outcomes due to infrastructure and resources. Traditional teaching methods were considered less effective compared to AI-based approaches.

3. Axis ٩: Impact of Parents' Involvement in Using AI Technologies on Language Development.

Table 9: Responses to Items Related to Axis 3 (Parental Role).

Nu.	Statement	Agree (%)	Some-what Agree (%)	Disagree (%)
29	Parents do not influence their children's learning using AI technologies.	10%	15%	75%
30	Parents influence their children's learning using AI technologies.	70%	20%	10%
31	Parents are applying AI technologies to assist teachers.	60%	25%	15%
32	Parents play a significant role in developing students' language communication skills after using AI technologies.	75%	20%	5%
33	Specialists play a significant role in developing students' language communication skills after using AI technologies.	65%	25%	10%

34	Teachers complement parents in developing students' language communication skills after using AI technologies.	80%	15%	5%
35	Students' language skills have improved after using AI technologies.	85%	10%	5%
36	Academic lifestyles do not develop the abilities of autistic students.	20%	25%	55%
37	Electronic media increase students' vocabulary development.	80%	15%	5%
38	Electronic media increase students' feelings of loneliness after use.	30%	25%	45%
39	The school's study supported student development alongside AI technologies. Students' positive evaluation rates increased after the use of AI technologies.	85%	10%	5%
40	The school's study supported student development alongside AI technologies. Students' positive evaluation rates increased after the use of AI technologies.	85%	10%	5%

Parents were widely perceived as playing a key role in supporting AI-assisted learning at home. Most agreed that parental involvement, along with teacher support, contributes significantly to language development.

Electronic media were seen as beneficial for vocabulary growth, although some concerns were raised about potential increases in loneliness. Schools were reported to have successfully integrated AI into the curriculum, leading to higher student engagement and positive evaluations

- ✓ Axis 1 (Gender and Age): AI significantly improves social and communication skills, especially among older children. Both genders benefit, though slightly more agreement exists for females being more affected.
- ✓ Axis 2 (School Type & Academic Year): Private and international schools showed greater success in implementing AI. Students' academic performance and digital literacy improved, and AI was seen as more effective than traditional methods.
- ✓ Axis 3 (Parental Involvement): Parents play a vital role in reinforcing AI-based learning at home. Collaboration between parents and teachers enhances language development outcomes. While electronic media boosts vocabulary, there is a moderate concern about its emotional effects.

Table 10: Validity and Reliability of the Study Axes.

Axis Number	Axis Title	No. of Items	Cronbach's Alpha	Internal Consistency Interpretation
1	Effectiveness of AI in developing social and linguistic skills by gender & age	14	0.87	High
2	Effectiveness of AI according to academic year and school type	14	0.83	High
3	Parental influence on linguistic development after using AI	12	0.89	Very High

The validity of the questionnaire was established through expert review, where specialists in special education and applied psychology confirmed the relevance and clarity of items related to each axis. Content validity ensured that all questions accurately reflected the study objectives and variables.

Reliability was assessed using Cronbach's Alpha , a widely used measure of internal consistency. The results show high reliability across all three axes:

➤ Axis 1 (Gender and Age): Cronbach's Alpha = 0.87, indicating strong internal consistency among the 14 items assessing the impact of AI on social and communication skills.

➤ Axis 2 (Academic Year and School Type): Cronbach's Alpha = 0.83, which reflects high reliability and suggests that the items measuring effectiveness based on educational context are consistent and cohesive.

➤ Axis 3 (Parental Involvement): Cronbach's Alpha = 0.89, demonstrating very high internal consistency, likely due to the clear thematic focus on parental roles and language development.

144 These values exceed the commonly accepted threshold of 0.70 for Cronbach's Alpha, confirming that the questionnaire is both valid and reliable for measuring the intended constructs among autistic children. This ensures the accuracy and trustworthiness of the findings derived from the collected data.

Table11: Hypotheses Testing Results.

Hypothesis Number	Hypothesis Statement	Statistical Test Used	Significance Level (p-value)	Decision
H1	There are no statistically significant differences in the development of social and communication skills between children in the experimental group and the control group after using AI techniques.	Independent Samples t-test	$p = 0.001$	Rejected
H2	There are statistically significant differences in favor of the experimental group in social and communication skills after using AI techniques.	Independent Samples t-test	$p = 0.001$	Supported
H3	There are statistically significant differences in the use of AI techniques in developing social and communication skills among autistic children according to the variables (gender and age).	One-Way ANOVA	Age: $p = 0.028$; Gender: $p = 0.267$	Partially Supported (Age significant, Gender not)

H4	There are no statistically significant differences in the effect of parental education on the development of social and linguistic communication skills in children with autism using AI techniques.	Chi-Square Test	$p = 0.062$	Accepted
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1) H1 Rejected : There is a significant difference between the experimental and control groups, indicating that AI had a positive impact.

2) H2 Supported : The experimental group showed significantly better outcomes, confirming the effectiveness of AI.

3) H3 Partially Supported : Age had a significant effect, but gender did not.

4) H4 Accepted : Parental education level did not significantly affect skill development.

Results.

1) The findings of the study revealed that the use of artificial intelligence (AI) techniques significantly contributes to the development of social and linguistic communication skills among children with autism spectrum disorder (ASD).

The experimental group, which was exposed to AI-based interventions, showed considerable improvements compared to the control group, as indicated by statistical analysis using independent samples t-tests.

2) Age played a significant role in the effectiveness of AI tools, with older children (10–12 years) demonstrating greater progress than younger ones (8–10 years), based on one-way ANOVA results.

However, no statistically significant differences were found between male and female participants, indicating that both genders benefit equally from AI-assisted learning.

3) School type also influenced outcomes, with students in private and international schools reporting higher levels of improvement due to better access to technology and trained personnel.

Parental education level did not show a significant impact on skill development; however, most parents reported being actively involved in supporting their children's use of AI at home.

4) Moreover, the questionnaire demonstrated high reliability across all three axes, with Cronbach's Alpha values exceeding 0.83, confirming its validity and consistency in measuring the intended constructs.

14 Discussion.

The findings of this study demonstrate that artificial intelligence (AI) techniques have a significant positive impact on the development of social and linguistic communication skills among children with Autism Spectrum Disorder (ASD).

The experimental group, which received AI-based interventions, showed statistically significant improvements compared to the control group, as evidenced by independent samples t-tests. This aligns with previous research that emphasizes the effectiveness of technology-assisted interventions in special education, particularly for children with developmental disorders like autism.

AI Techniques Used and Their Role in Achieving Results.

Several AI-based tools and technologies were integrated into the intervention program, including:

- 1) Speech Recognition Software : Enabled real-time feedback on pronunciation and sentence structure, helping children improve expressive language skills.
- 2) Interactive Virtual Learning Platforms : Offered structured scenarios where children could practice initiating conversations, responding to questions, and interpreting facial expressions.

3) Adaptive Learning Applications : Tailored content based on individual progress, ensuring personalized learning paths suited to each child's pace and ability.

4) AI-Powered Chatbots and Social Robots : Simulated social interactions and encouraged peer-like engagement, promoting emotional recognition and turn-taking in conversation.

5) Gamified Language Exercises : Motivated students through reward systems and interactive storytelling, increasing engagement and retention.

144 These technologies provided structured, repetitive, and engaging environments , which are essential for children with ASD who often benefit from predictable and visually supported learning experiences.

The use of immediate feedback mechanisms and adaptive difficulty levels ensured that children remained motivated and challenged at appropriate levels throughout the intervention.

Age Differences and Developmental Readiness.

The study revealed that older children (10–12 years) demonstrated significantly greater improvement than younger ones (8–10 years), as indicated by one-way ANOVA results.

This may be attributed to increased cognitive maturity , better digital literacy , and improved attention span , which allowed older children to engage more effectively with complex AI interfaces. These findings support those of Chuan-Po Wang (2023) , who reported similar age-related benefits when using AI robots for sentence organization training in children with ASD.

However, this contrasts with the findings of Al Sharkawy & Al Hattalia (2024) , who found no significant differences in language skill improvement across age groups when using AI applications.

The discrepancy may stem from differences in intervention duration , type of AI tool used , or sample characteristics .

Gender and Inclusivity of AI Interventions.

No statistically significant differences were observed between male and female participants, indicating that both genders benefited equally from AI-assisted learning. This is a crucial finding, especially considering the higher prevalence of ASD in males.

It suggests that AI-based educational strategies can be effectively generalized across genders, offering equal opportunities for skill development regardless of sex. These results are consistent with those of Kotsi et al. (2025), who also found no gender disparities in AI intervention outcomes among children with ASD.

School Type and Infrastructure Influence.

Students attending private and international schools showed greater improvement in social and linguistic skills compared to those in government-run schools.

This outcome highlights the critical role of infrastructure , availability of trained personnel , and classroom environment in the successful implementation of AI-based interventions.

Private schools often provide smaller class sizes , better access to digital tools , and more flexible curricula , which facilitate effective integration of AI into teaching practices.

This finding supports the conclusions of Gitimoghaddam et al. (2022) , who emphasized the importance of institutional readiness in adopting assistive technologies.

1441 However, it contrasts with the findings of Du et al. (2024) , who argued that even basic AI tools can yield significant improvements if properly implemented in public school settings. This suggests that with adequate support , government schools can also benefit from AI interventions.

Parental Involvement and Home Reinforcement.

Although parental education level did not show a statistically significant impact on skill development, most parents-regardless of academic background-reported being actively involved in supporting their children's use of AI at home.

This highlights the importance of parental engagement as a complementary factor in reinforcing classroom learning. Parents facilitated consistent practice , motivation , and emotional support , which enhanced the overall effectiveness of the intervention.

1441 This aligns with the findings of Li et al. (2024) , who emphasized the value of home-school collaboration in maximizing the benefits of AI-powered interventions for children with ASD.

Comparison with Previous Studies.

The current study corroborates earlier research that has documented the efficacy of AI in enhancing communication and social skills in children with autism, such as Belpaeme et al. (2013) and Esteban et al. (2017) , who highlighted the potential of robot-assisted therapy and adaptive learning systems.

However, unlike many previous studies that focused on small sample sizes or single-case designs, this study involved a larger, diverse sample and employed quantitative analysis , strengthening the validity of its conclusions.

One notable difference is that parental education level did not significantly influence outcomes in this study, contradicting some earlier literature that suggested higher-educated parents might provide more effective home reinforcement.

This could indicate that motivation and involvement are more important than formal education when it comes to supporting children's use of AI tools.

Conclusion.

In summary, this study confirms that AI-based interventions can significantly enhance social and linguistic communication skills in children with ASD. The integration of specific AI tools , such as speech recognition software , interactive platforms , and adaptive learning systems , played a central role in achieving these outcomes.

The results are largely consistent with previous research, although variations in findings underscore the importance of contextual factors , such as age , school type , and parental involvement , in determining the success of AI-based educational strategies.

Future research should focus on longitudinal assessments of AI interventions, customizing AI tools for different severity levels of ASD , and expanding access to under-resourced schools to ensure equitable benefits for all children.

Recommendations.

Based on the results and discussion, the following recommendations are proposed:

- 1) Integrate AI tools into special education curricula , particularly for developing social and linguistic skills in children with autism.
- 2) Provide professional development programs for teachers to enhance their ability to implement AI-based interventions effectively.
- 3) Promote parental awareness and involvement in using AI technologies at home through workshops and guidance materials.
- 4) Ensure equitable access to AI resources across all school types , including government-run institutions, to reduce disparities in educational outcomes.

- 5) Encourage early exposure to AI-based learning tools starting at a young age to maximize developmental benefits.
- 6) Conduct longitudinal studies to assess the long-term impact of AI interventions on children with ASD.
- 7) Develop culturally appropriate AI content that aligns with local educational standards and linguistic contexts.

Suggested Research:

- 1) Scientific research should be conducted on language communication skills and how to develop them in families of autistic children.
- 2) Scientific studies should be conducted to support the development of language skills in children with autism in social institutions.
- 3) Scientific studies should be conducted to support social skills in children with autism and how to train families and social institutions, such as schools, to develop these skills.

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